UPSSSC JE

MECHANICAL
FLUID MECHANICS & HYDRAULIC MACHINES

Complete English Edition

Unit 8 · 10 Marks · 2026

Selection of hydraulic turbines — by head

Pelton (impulse)	High head > 250 m	Ns 8.5–51
Francis (reaction, mixed flow)	Medium head 60–250 m	Ns 51–255
Kaplan (reaction, axial)	Low head < 60 m	Ns 255–860

$$N_s = N\sqrt{P} / H^{5/4}$$

UPSSSC JE (Mechanical) 2026: Part-1 has **10 questions (10 marks)** from **Fluid Mechanics and Hydraulic Machines**. Syllabus: fluid statics, measurement of fluid pressure, hydraulic turbines and centrifugal pumps — classification, principles and performance.

Each chapter contains key points, diagrams, formula table, detailed explanation, solved numericals, 30 one-liners, MCQs with explanations and SSC JE PYQs. Content follows standard BTEUP diploma textbooks (R. K. Bansal, R. K. Rajput, Modi–Seth).

For a better score, study this PDF and also attempt the QUIZ.

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Chapter 1: Fluid Properties & Fluid Statics

Fluid Properties & Fluid Statics – Basic Concept

A **fluid** is a substance that deforms continuously under even the smallest shear force — both liquids and gases. **Fluid statics** studies fluids at rest: pressure, its distribution, forces on submerged surfaces and buoyancy. Dams, tanks, ships and hydraulic presses are designed on it.

Understand in Detail — Fluid Properties & Fluid Statics

1.1 What is a fluid — how is it different from a solid?

Place a brick on a table and push its top gently sideways — it deforms slightly and stops. Now push the surface of water gently sideways with your hand — the water keeps on flowing. This difference is the definition of a fluid: **a fluid is a substance that deforms (flows) continuously under the action of even the smallest shear stress.** Fluids include both liquids and gases.

Density (ρ) = mass/volume (kg/m³); water 1000 kg/m³. **Specific weight (w = ρg)** = weight/volume (N/m³); water 9810 N/m³. **Specific gravity (S)** = density of fluid ÷ density of water (at 4°C) — it has no unit; mercury 13.6. **Specific volume** = 1/ρ.

1.2 Viscosity — the 'internal friction' of a fluid

Honey flows slowly and water fast — because honey has higher **viscosity**. Viscosity is the property of a fluid that resists the sliding of its layers over one another. Imagine the fluid is made of thin layers like a pack of cards; when the top layer is moved, the lower layers drag behind due to friction.

Suppose fluid fills the gap between two parallel plates, the lower plate fixed and the upper moving with velocity u. The fluid layer touching a plate moves with the plate (**no-slip condition**). So velocity varies from zero at the bottom to u at the top. The rate of this change, **du/dy**, is called the **velocity gradient**.

Derivation: Newton's law of viscosity

1. Experiments show that the shear stress τ is proportional to the velocity gradient: $\tau \propto du/dy$
2. $\tau = \mu (du/dy)$; the constant μ = **dynamic (absolute) viscosity**
3. Unit: $N\cdot s/m^2 = Pa\cdot s$ (SI) ; in CGS the **poise**, 1 poise = 0.1 $N\cdot s/m^2$
4. **Kinematic viscosity** $\nu = \mu/\rho$; unit m^2/s, in CGS the **stoke** = 1 $cm^2/s = 10^{-4}\ m^2/s$

Effect of temperature — a favourite exam question: in liquids viscosity comes from **cohesion** between molecules; as temperature rises molecules move apart, cohesion falls → **viscosity decreases** (hot oil is thinner). In gases viscosity comes from **momentum exchange** of molecules; as temperature rises molecules move faster, exchange increases → **viscosity increases**.

Types of fluids: ideal fluid — zero viscosity, incompressible (imaginary). **Newtonian** — graph of τ vs du/dy is a straight line through the origin (water, air, petrol). **Non-Newtonian** — curved graph (blood, paint). **Ideal plastic (Bingham)** — needs a minimum yield stress first, then linear (toothpaste, sewage sludge).

> **Example:** An oil film of 0.025 mm lies between two plates. The upper plate moves at 60 cm/s and needs a force of 2 N/m² per unit area. Find the viscosity of the oil.
>
> **Solution:**
>
> 1. du = 0.6 m/s, dy = 0.025 × 10^{-3} m → du/dy = 0.6/0.000025 = 24000 s^{-1}
> 2. μ = τ/(du/dy) = 2/24000 = **8.33 × 10^{-5} N·s/m²** = 8.33 × 10^{-4} poise

1.3 Surface tension, capillarity and compressibility

Why is a water drop round? And how can a mosquito stand on water? The reason is **surface tension (σ)**. A molecule inside a liquid is pulled equally in all directions, but a surface molecule only from below and the sides — so the surface behaves like a stretched membrane. σ = force per unit length of an imaginary line drawn on the surface, unit **N/m**. For water–air about 0.073 N/m.

Excess pressure inside: imagine cutting a drop in half — the excess pressure Δp acts on the cut circle area (πd²/4); it is balanced by surface tension acting along the circumference (πd). Δp × πd²/4 = σ × πd → **droplet: Δp = 4σ/d**. **Soap bubble** — two surfaces (inner and outer), so **Δp = 8σ/d**. **Liquid jet** (cylinder) — **Δp = 2σ/d**.

Capillarity: dip a thin glass tube in water and water rises in it; dip it in mercury and mercury is depressed. Water 'likes' glass (adhesion > cohesion, contact angle θ < 90°), mercury does not (θ ≈ 130°). Upward surface-tension force σ cosθ × πd = weight of risen column ρg × (πd²/4) × h → **h = 4σ cosθ/(ρ g d)**. The thinner the tube, the higher the rise. Water rising in plants and oil rising in a lamp wick are examples of capillarity.

Compressibility and bulk modulus (K): K = −dp/(dV/V) = increase in pressure ÷ volumetric strain. For water K ≈ 2.1 × 10^{9} N/m² — i.e. about 210 bar is needed to reduce the volume of water by 1%! That is why liquids are treated as **incompressible** in normal calculations. **Compressibility = 1/K**. **Vapour pressure:** the pressure at which a liquid boils at a given temperature; if anywhere in a pump or turbine the pressure falls below the vapour pressure, bubbles form — this is the root of **cavitation**.

> **Example:** How high will water rise in a 2 mm diameter glass tube? σ = 0.073 N/m, θ = 0°.
>
> **Solution:**
>
> 1. h = 4 × 0.073 × 1/(1000 × 9.81 × 0.002)
> 2. h = 0.292/19.62 = 0.01488 m = **14.9 mm**

1.4 Fluid statics — why does pressure increase with depth?

The deeper a diver goes, the more pressure he feels on his ears — because the column of water above him is taller. **Pressure** = normal force per unit area, p = F/A, unit Pa (N/m²); 1 bar = 10^{5} Pa.

Pascal's law: in a fluid at rest the pressure at a point is equal in all directions. A **hydraulic press/jack** works on this principle: the pressure $p = F_1/A_1$ applied on a small piston is transmitted undiminished through the fluid to the large piston, where the force $F_2 = p\,A_2 = F_1\,(A_2/A_1)$ — a small force lifts a large load.

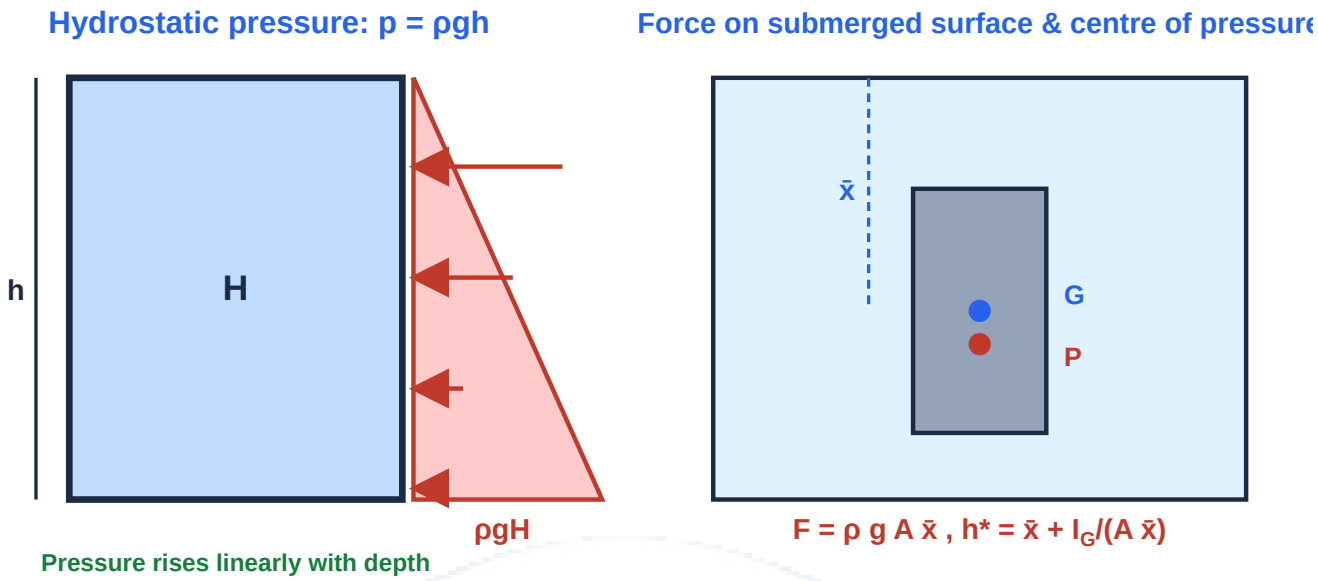


Fig 8.1: Hydrostatic pressure distribution and total pressure on a submerged surface

Derivation: Hydrostatic law p = ρgh

1. Take a small vertical cylindrical element in the fluid — height dz, cross-section dA
2. Upward force from below p dA, downward force from above (p + dp) dA, weight of element ρg dA dz
3. Equilibrium: $p\,dA - (p + dp)\,dA - \rho g\,dA\,dz = 0 \rightarrow$ **$dp/dz = -\rho g$**
4. Measuring depth h downward from the free surface gives **$p = \rho g h$** (gauge pressure)
5. So pressure depends only on depth, not on the shape of the vessel — the **hydrostatic paradox**: vessels of different shapes filled to the same height have the same pressure at the base

Pressure head: $h = p/(\rho g)$ — expressing a pressure as 'so many metres of fluid column'. 1 atm = 101.325 kPa = **10.33 m of water** = **760 mm of mercury**. **Absolute pressure = atmospheric + gauge pressure**; if the pressure is below atmospheric it is a **vacuum** pressure: absolute = atmospheric − vacuum.

1.5 Total pressure force and centre of pressure on a submerged surface

When designing a dam, a tank wall or a sluice gate, two questions are most important: (1) how much total force does the water exert? (2) at which point does this force act? The answer to the second is the **centre of pressure**. Since pressure increases with depth, the pressure distribution on a wall is **triangular** — zero at the top and maximum at the bottom.

Derivation: Total pressure force and depth of centre of pressure

1. Divide the surface into thin horizontal strips; force on strip dA at depth h is dF = ρgh dA
2. Total force F = ρg ∫h dA = ρg A $\bar{h}$ (since ∫h dA = A $\bar{h}$, $\bar{h}$ = depth of centroid) → **F = ρ g A $\bar{h}$**
3. Moment about the free surface: F × h* = ρg ∫h² dA = ρg I_0, where I_0 is the moment of inertia about the free surface
4. Parallel axis theorem: $I_0 = I_G + A\bar{h}^2$ → **h* = $\bar{h}$ + I_G/(A $\bar{h}$)** (for a vertical surface)
5. Since I_G/(A $\bar{h}$) is always positive, the **centre of pressure always lies below the centroid**; for an inclined surface (angle θ) h* = $\bar{h}$ + I_G sin²θ/(A $\bar{h}$)

A shortcut that helps a lot: for a vertical rectangular wall whose top edge is at the water surface — the centroid of the pressure triangle lies at 1/3 height from the bottom, so the **centre of pressure is at depth 2H/3** from the surface. And total force = ρg(H/2)(B H) = ρg B H²/2.

Buoyancy and stability: Archimedes — upward force on an immersed body = weight of fluid displaced; it acts at the centroid of the displaced volume, the **centre of buoyancy (B)**. When a floating body tilts slightly, B shifts; the point where the new buoyant force line cuts the axis is the **metacentre (M)**. **M above the centre of gravity G → stable** (rights itself); M below G → unstable (capsizes); M = G → neutral. **Metacentric height GM = I/V − BG**. In ships a very large GM gives fast, uncomfortable rolling; a very small GM risks capsizing.

Example: A rectangular gate 2 m wide and 3 m high stands vertically with its top edge 1 m below the water surface. Find the total force and the centre of pressure.

Solution:

1. A = 2 × 3 = 6 m² ; $\bar{h}$ = 1 + 3/2 = 2.5 m
2. F = 1000 × 9.81 × 6 × 2.5 = 147150 N = **147.2 kN**
3. I_G = b d³/12 = 2 × 27/12 = 4.5 m⁴
4. h* = 2.5 + 4.5/(6 × 2.5) = 2.5 + 0.3 = **2.8 m** — 0.3 m below the centroid ✔

1. Key Points

- **Density and specific weight:** ρ = m/V (water 1000 kg/m³); specific weight w = ρg (water 9810 N/m³); specific gravity S = ρ/ρ_{water} (mercury 13.6).
- **Viscosity:** Newton's law of viscosity τ = μ du/dy; dynamic viscosity μ (N·s/m² = Pa·s; 1 poise = 0.1 Pa·s); kinematic viscosity ν = μ/ρ (m²/s; 1 stoke = 10^{-4} m²/s).
- **Effect of temperature:** viscosity of liquids decreases with temperature (cohesion falls), while that of gases increases (molecular momentum exchange).
- **Surface tension and capillarity:** σ (N/m); excess pressure in a droplet Δp = 4σ/d, soap bubble 8σ/d, jet 2σ/d; capillary rise h = 4σ cos θ/(ρgd) — rise in water, fall in mercury.
- **Pascal's law:** pressure at a point in a fluid at rest is equal in all directions; hydraulic presses, jacks and brakes are based on it.

- **Hydrostatic law:** $dp/dz = -\rho g$; pressure at depth h, $p = \rho g h$ (gauge). Pressure does not depend on the shape of the vessel (hydrostatic paradox).
- **Total pressure on a submerged surface:** $F = \rho g A \bar{x}$ ($\bar{x}$ = depth of centroid). Depth of centre of pressure $h^* = \bar{x} + I_G/(A\bar{x})$ — always below the centroid.
- **Buoyancy:** Archimedes — buoyant force = weight of displaced fluid; it acts at the centre of buoyancy (centroid of displaced volume).
- **Stability of floating bodies:** metacentre (M) above G — stable; below — unstable; coincident — neutral. Metacentric height $GM = I/V - BG$.
- **Units of pressure:** 1 Pa = 1 N/m²; 1 bar = 10^5 Pa; 1 atm = 101.325 kPa = 760 mm Hg = 10.33 m of water.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Density / specific weight	**$\rho = m/V$, $w = \rho g$**
Newton's law of viscosity	**$\tau = \mu\, du/dy$**
Kinematic viscosity	**$\nu = \mu/\rho$**
Droplet / bubble / jet	**$4\sigma/d$, $8\sigma/d$, $2\sigma/d$**
Capillary rise	**$h = 4\sigma \cos\theta/(\rho g d)$**
Hydrostatic pressure	**$p = \rho g h$**
Total pressure force	**$F = \rho g A \bar{x}$**
Centre of pressure	**$h^* = \bar{x} + I_G/(A\bar{x})$**
Buoyant force	**$F_B = \rho g V_{displaced}$**
Metacentric height	**$GM = I/V - BG$**
Absolute pressure	**$p_{abs} = p_{atm} + p_{gauge}$**

3. Detailed Explanation

- **Newtonian and non-Newtonian fluids:** Newtonian (water, air, oil) have $\tau \propto du/dy$; non-Newtonian do not — plastic/Bingham (toothpaste), pseudo-plastic (blood, paint), dilatant (starch solution).
- **Ideal fluid:** incompressible with zero viscosity — not real, an assumption for analysis.
- **Inclined/vertical surfaces:** for a vertical rectangular gate with top at the free surface, the centre of pressure is at 2/3 of the depth (triangular pressure distribution).
- **Link to fluid dynamics — Bernoulli:** $p/\rho g + V^2/2g + z$ = constant (ideal, steady, incompressible, along a streamline) — used in venturimeters, Pitot tubes, turbines and pumps; continuity $A_1V_1 = A_2V_2$.
- **Hydraulic press:** $F/a = W/A$, so $W = F \times A/a$ — a small force lifts a heavy load; mechanical advantage A/a.

4. Comparison Table — Viscosity of Liquids vs Gases

Basis	Liquids	Gases
Main cause	Cohesive forces	Molecular momentum exchange
With rising temperature	Decreases	Increases
Magnitude	Higher	Much lower

4. Comparison Table — Stability of Floating Body

Condition	Position	Result
Stable	M above G (GM > 0)	Returns when tilted
Unstable	M below G (GM < 0)	Capsizes
Neutral	M = G	Stays in new position

5. Solved Numerical Problems

Example 1. Find the pressure at 5 m depth in water (g = 9.81).
Solution:

1. $p = 1000 \times 9.81 \times 5 =$ **49.05 kPa** (gauge)

Example 2. A vertical rectangular gate 1 m wide × 2 m deep has its top at the water surface. Find total pressure and centre of pressure.
Solution:

1. $\bar{x} = 1$ m, $A = 2$ m²
2. $F = 1000 \times 9.81 \times 2 \times 1 =$ **19.62 kN**
3. $I_G = 1 \times 2^3/12 = 0.667$ m⁴; $h^* = 1 + 0.667/(2 \times 1) =$ **1.333 m** (2/3 depth)

Example 3. Find the capillary rise of water in a 0.5 mm glass tube ($\sigma = 0.073$ N/m, $\theta = 0°$).
Solution:

1. $h = 4 \times 0.073/(1000 \times 9.81 \times 0.0005) =$ **59.5 mm**

Example 4. A hydraulic press has plungers of 3 cm and 30 cm diameter. Find the load lifted by 100 N.
Solution:

1. $W = 100 \times (30/3)^2 =$ **10,000 N**

Example 5. Oil ($\mu = 0.1$ Pa·s) is between plates 2 mm apart; the top plate moves at 1 m/s. Find the shear stress.
Solution:

1. $\tau = \mu\, du/dy = 0.1 \times 1/0.002 =$ **50 N/m²**

6. Frequently Asked Facts

- Viscosity of liquids falls with temperature; of gases it rises.
- Centre of pressure is always below the centroid; at 2/3 depth for a vertical rectangle up to the surface.
- GM > 0 is stable; 1 atm = 10.33 m of water = 760 mm Hg.
- Excess pressure in a soap bubble is 8σ/d (two surfaces).

7. Note (Confusing Facts)

- Mercury shows capillary fall because $\theta > 90°$.
- Pressure depends only on depth, not on the shape of the vessel.
- The unit of kinematic viscosity is the stoke, not the poise.

Remember: One line: $p = \rho gh$; $F = \rho g A\bar{x}$. Assertion–Reason: (A) Dam walls are thicker at the bottom. (R) Water pressure increases with depth. — Both true; R explains A.

30 Most Important One-Liner Facts — Fluid Properties & Fluid Statics

1. Density of water?

— 1000 kg/m³.

2. Specific weight of water?

— 9810 N/m³.

3. Specific gravity of mercury?

— 13.6.

4. Newton's law of viscosity?

— $\tau = \mu \, du/dy$.

5. SI unit of dynamic viscosity?

— Pa·s (N·s/m²).

6. 1 poise = ?

— 0.1 Pa·s.

7. Kinematic viscosity?

— $\nu = \mu/\rho$.

8. 1 stoke = ?

— 10^{-4} m²/s.

9. Liquid viscosity with temperature?

— Decreases.

10. Gas viscosity with temperature?

— Increases.

11. Unit of surface tension?

— N/m.

12. Excess pressure in droplet?

— 4σ/d.

13. Excess pressure in soap bubble?

— 8σ/d.

14. Capillary rise?

— $4\sigma \cos\theta/(\rho g d)$.

15. Capillarity in mercury?

— Fall.

16. Pascal's law?

— Pressure equal in all directions.

17. Based on Pascal's law?

— Hydraulic press, jack, brakes.

18. Pressure at depth h?

— ρgh.

19. Total force on submerged surface?

— $\rho g A\bar{x}$.

20. Centre of pressure?

— $\bar{x} + I_G/(A\bar{x})$.

21. Position of centre of pressure?

— Below centroid.

22. Archimedes' principle?

— Buoyant force = weight of displaced fluid.

23. Centre of buoyancy?

— Centroid of displaced volume.

24. Stable floating body?

— M above G.

25. Metacentric height?

— GM = I/V − BG.

26. 1 atm = ? m of water

— 10.33.

27. 1 atm = ? mm Hg

— 760.

28. Absolute pressure?

— Atmospheric + gauge.

29. Zero viscosity, incompressible fluid?

— Ideal fluid.

30. Toothpaste is which fluid?

— Bingham plastic.

Practice MCQs — Fluid Properties & Fluid Statics

Q1. Viscosity of liquids with rising temperature:

(a) Increases

(b) Decreases

(c) Constant

(d) First rises then falls

✔ **Answer: (b) Decreases**

Explanation: Cohesive forces fall.

Q2. Unit of kinematic viscosity:

(a) Pa·s

(b) m^2/s

(c) N/m

(d) kg/m^3

✔ **Answer: (b) m^2/s**

Explanation: $\nu = \mu/\rho$.

Q3. Excess pressure inside a soap bubble:

(a) $2\sigma/d$

(b) $4\sigma/d$

(c) $8\sigma/d$

(d) σ/d

✔ **Answer: (c) $8\sigma/d$**

Explanation: Two surfaces.

Q4. The centre of pressure lies:

(a) Above the centroid

(b) Below the centroid

(c) At the centroid

(d) At the surface

✔ **Answer: (b) Below the centroid**

Explanation: $I_G/(A\bar{x}) > 0$.

Q5. A floating body is stable when:

(a) M is below G

(b) M is above G

(c) M = G

(d) B is above G

✔ **Answer: (b) M is above G**

Explanation: GM > 0.

Q6. 1 atm pressure equals:

(a) 10.33 m of water
(b) 1 m of water
(c) 76 mm Hg
(d) Exactly 1 bar

✔ **Answer: (a) 10.33 m of water**

Explanation: 760 mm Hg = 10.33 m water.

Q7. A hydraulic press works on:

(a) Archimedes
(b) Pascal
(c) Bernoulli
(d) Newton

✔ **Answer: (b) Pascal**

Explanation: Pressure transmitted equally.

Q8. Capillary effect in mercury:

(a) Rise
(b) Fall
(c) None
(d) Both

✔ **Answer: (b) Fall**

Explanation: $\theta > 90°$.

Q9. Pressure at 5 m depth of water (g = 9.81):

(a) 4.905 kPa
(b) 49.05 kPa
(c) 490.5 kPa
(d) 5 kPa

✔ **Answer: (b) 49.05 kPa**

Explanation: $1000 \times 9.81 \times 5$.

Q10. 1 poise =

(a) 1 Pa·s
(b) 0.1 Pa·s
(c) 10 Pa·s
(d) 0.01 Pa·s

✔ **Answer: (b) 0.1 Pa·s**

Explanation: 0.1 $N \cdot s/m^2$.

Q11. Vertical rectangle with top at surface — depth of centre of pressure:

(a) d/2
(b) 2d/3
(c) d/3
(d) 3d/4

✔ **Answer: (b) 2d/3**

Explanation: Triangular distribution.

Q12. An ideal fluid is:

(a) Incompressible, zero viscosity
(b) Compressible, viscous
(c) Newtonian
(d) Bingham

✔ **Answer: (a) Incompressible, zero viscosity**

Explanation: Hypothetical.

Q13. Buoyant force equals:

(a) Weight of body
(b) Weight of displaced fluid
(c) Volume of body
(d) Pressure × area

✔ **Answer: (b) Weight of displaced fluid**

Explanation: Archimedes.

Q14. Hydraulic press, plungers 3 cm and 30 cm, 100 N — load:

(a) 1000 N
(b) 10,000 N
(c) 100 N
(d) 1,00,000 N

✔ **Answer: (b) 10,000 N**

Explanation: 100 × 100.

Q15. Blood is which type of fluid?

(a) Newtonian
(b) Pseudo-plastic
(c) Dilatant
(d) Ideal

✔ **Answer: (b) Pseudo-plastic**

Explanation: Shear-thinning.

SSC JE / UPSSSC JE Previous Year Questions — Fluid Properties & Fluid Statics

Q1. Unit of surface tension:

SSC JE (Previous Year)

(a) N/m^2
(b) N/m
(c) N·m
(d) N

✔ **Answer: (b) N/m**

Explanation: Force per length.

Q2. Viscosity of gases with temperature:

SSC JE (Previous Year)

(a) Decreases
(b) Increases
(c) Constant
(d) Zero

✔ **Answer: (b) Increases**

Explanation: Molecular momentum exchange.

Q3. For stability of a floating body, the metacentre should be:

SSC JE (Previous Year)

(a) Above the centre of gravity
(b) Below
(c) Coincident
(d) At centre of buoyancy

✔ **Answer: (a) Above the centre of gravity**

Explanation: GM > 0.

Q4. Total pressure on a submerged plane surface:

SSC JE (Previous Year)

(a) ρgA

(b) ρgA$\bar{x}$

(c) ρg$\bar{x}$

(d) ρA$\bar{x}$

✔ **Answer: (b) ρgA$\bar{x}$**

Explanation: F = ρgA$\bar{x}$.

Q5. Newton's law of viscosity relates:

SSC JE (Previous Year)

(a) Pressure and velocity

(b) Shear stress and velocity gradient

(c) Density and pressure

(d) Temperature and viscosity

✔ **Answer: (b) Shear stress and velocity gradient**

Explanation: $\tau = \mu \, du/dy$.

Q6. Capillary rise varies with diameter as:

SSC JE (Previous Year)

(a) Directly proportional

(b) Inversely proportional

(c) Proportional to square

(d) Independent

✔ **Answer: (b) Inversely proportional**

Explanation: $h \propto 1/d$.

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Chapter 2: Measurement of Fluid Pressure

Measurement of Fluid Pressure – Basic Concept

Measuring pressure is essential in industry, pipelines, boilers and hydraulic plants. Pressure-measuring devices are of two types — **manometers** (based on balancing a liquid column) and **mechanical gauges** (based on deformation of an elastic element, e.g. the Bourdon tube).

Understand in Detail — Measurement of Fluid Pressure

2.1 Pressure measurement — which pressure, from which reference?

Before measuring pressure it is important to understand from which 'zero' we measure. **Absolute pressure** is measured from a perfect vacuum (zero). **Gauge pressure** is measured from the local atmospheric pressure — most gauges show this; '30 psi' in a tyre is gauge pressure. **Vacuum pressure** — how much below atmospheric.

Barometer: measures atmospheric pressure. Fill a long tube with mercury and invert it in a mercury dish — mercury stops at about 760 mm and a **Torricellian vacuum** forms above. Mercury is used because it is heavy (a water barometer would need a 10.33 m tube!) and its vapour pressure is negligible.

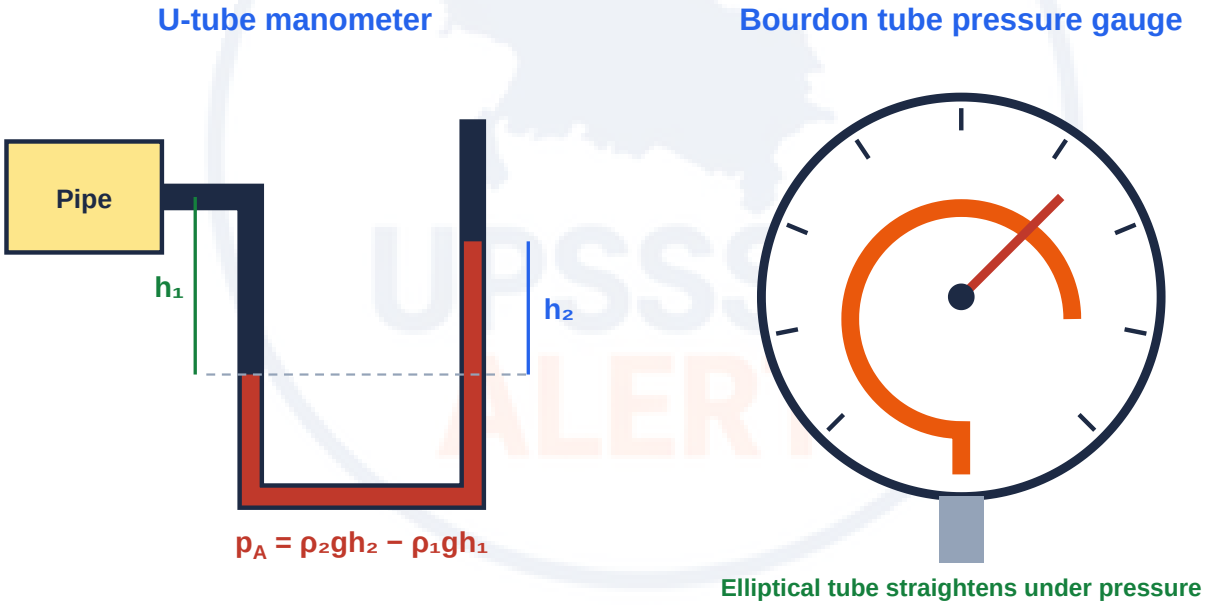


Fig 8.2: U-tube manometer and Bourdon tube pressure gauge

2.2 Manometers — reading pressure from a liquid column

Piezometer: the simplest — attach a vertical transparent tube to the pipe; the pipe pressure raises the liquid to height h in the tube, p = ρgh. **Limitations:** (1) needs a very long tube for high pressure, (2)

cannot measure gas pressure (the gas escapes), (3) cannot measure negative (vacuum) pressure (air gets sucked in).

U-tube manometer: a U-shaped tube filled with a heavy liquid (usually mercury); one limb is connected to the pipe and the other open to the atmosphere. All three limitations are removed — the heavy liquid keeps the tube short, and both gas and vacuum can be measured.

The 'golden rule' for solving (step by step): (1) start at point A in the pipe and write p_A. (2) move along the tube — **going down add +ρgh, going up subtract −ρgh** (using that part's ρ). (3) two points at the same horizontal level in the same continuous liquid have equal pressure. (4) on reaching the open end set it equal to 0 (gauge) or to the pressure of the other point. That's all — every manometer question is solved this way.

Differential manometer: measures the **difference** of pressure between two pipes or two points of a pipe — as in a venturimeter. If both sides carry the same liquid (S_1) and the manometric liquid is S_m, at the same level: $\mathbf{p_A - p_B = \rho g\ x\ (S_m - S_1)}$ (ρ = density of water, x = difference of mercury levels). For mercury–water, 1 m of mercury difference = 12.6 m of water head. **Inverted U-tube** — for very small differences, light fluid (air/oil) on top. **Micro-manometer (inclined tube)** — tilting the tube spreads a small vertical difference over a long reading, increasing sensitivity.

> **Example:** A U-tube manometer is connected to a water pipe. The pipe centre is 0.2 m above the left mercury level; in the right (open) limb mercury stands 0.3 m above the left level. Pressure in the pipe?
> **Solution:**
> 1. From p_A down 0.2 m in water: $p_A + 1000 \times 9.81 \times 0.2$
> 2. In the right limb mercury rises 0.3 m above that level to the open end: $= 0 + 13600 \times 9.81 \times 0.3$
> 3. $p_A = 40025 - 1962 = 38063$ Pa ≈ **38.06 kPa (gauge)**

2.3 Mechanical gauges — Bourdon tube and others

Measuring high pressures (e.g. 20 bar in a boiler) with a manometer is impractical — the mercury column alone would be 15 m long! In such cases **mechanical gauges** are used.

Bourdon tube gauge: inside is a metal tube of **elliptical cross-section** bent into a C-shape. One end is fixed and connected to the pressure source, the other is closed and free. As pressure rises inside, the ellipse tries to become round and the tube **tries to straighten**; the free end moves and, through a link, sector and pinion, this is converted into rotation of the dial pointer. It shows **gauge pressure** and can also measure vacuum (the tube then curls more).

Diaphragm gauge — a thin flexible membrane bulges under pressure; for low pressures. **Bellows gauge** — an accordion-like metal bellows. **Dead-weight tester** — produces an accurate pressure with a piston of known weight and area, used to **calibrate** other gauges.

> **Note:** Remember: the Bourdon tube's section is **elliptical**, not circular — a circular tube would not straighten under pressure. And a gauge always shows gauge pressure, so do not forget to add atmospheric pressure to get absolute.

1. Key Points

- **Types of pressure:** atmospheric (101.325 kPa at sea level), gauge pressure (relative to atmosphere), vacuum/negative gauge (below atmosphere), absolute pressure (relative to perfect vacuum).
- **Measuring atmospheric pressure:** barometer — mercury barometer (760 mm Hg) and aneroid barometer.
- **Piezometer:** the simplest manometer — a vertical tube attached to the pipe; only for small positive pressures of liquids; cannot measure gases or negative pressure.
- **Simple U-tube manometer:** with a heavy manometric liquid (mercury) both high and negative pressures can be measured; $p_A = \rho_2 g h_2 - \rho_1 g h_1$.
- **Differential manometer:** pressure difference between two points; U-tube differential: $p_A - p_B = gh(\rho_g - \rho)$ (same fluid, same level); inverted U-tube (lighter liquid) for small differences.
- **Micromanometer and inclined manometer:** for very small pressure differences; in an inclined tube the reading L gives height $h = L \sin\theta$ — sensitivity increases.
- **Bourdon tube pressure gauge:** a curved metal tube of elliptical section tends to straighten under pressure; motion goes through gear–sector to the pointer; most widely used for high pressure — reads gauge pressure only.
- **Other mechanical gauges:** diaphragm gauge (low pressure), bellows gauge, dead-weight tester (for calibrating gauges).
- **Properties of manometric liquid:** low vapour pressure, immiscible, clear meniscus, suitable density; mercury (heavy), water, alcohol, kerosene (light).

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Absolute pressure	$p_{abs} = p_{atm} + p_{gauge}$
Vacuum pressure	$p_{abs} = p_{atm} - p_{vac}$
Piezometer	$p = \rho g h$
Simple U-tube	$p_A = \rho_2 g h_2 - \rho_1 g h_1$
U-tube (negative pressure)	$p_A = -(\rho_1 g h_1 + \rho_2 g h_2)$
Differential U-tube	$p_A - p_B = g h (\rho_g - \rho)$
Inclined manometer	$h = L \sin\theta$
Pressure head	$h = p/(\rho g)$
Column conversion	$h_1\rho_1 = h_2\rho_2$ $(S_1h_1 = S_2h_2)$

3. Detailed Explanation

- **Method for manometers:** start at one point; add ρgh going down and subtract going up; equate to the pressure at the other end. Pressure is equal at the same horizontal level in the same continuous

liquid.

- **Why mercury:** high density (13.6) gives a short column for large pressure, low evaporation and immiscible with water; but it is toxic.
- **Gauge readings:** a Bourdon gauge shows gauge pressure; add atmospheric pressure for absolute. A vacuum gauge shows the amount below atmospheric.
- **Venturimeter and Pitot tube:** in flow measurement, the pressure difference read on a differential manometer gives velocity/flow through Bernoulli; a Pitot tube reads stagnation minus static pressure: $V = \sqrt{(2gh)}$.

4. Comparison Table — Manometer vs Mechanical Gauge

Basis	Manometer	Mechanical gauge (Bourdon)
Principle	Liquid column balance	Elastic deformation
Range	Low to medium	Up to high
Accuracy	High (standard)	Depends on calibration
Portability	Low	High

5. Solved Numerical Problems

Example 1. A mercury U-tube manometer is connected to a water pipe; $h_1 = 0.2$ m (water), $h_2 = 0.3$ m (mercury). Find the pipe pressure.

Solution:

1. $p_A = 13600 \times 9.81 \times 0.3 - 1000 \times 9.81 \times 0.2$
2. $= 40{,}025 - 1{,}962 =$ **38.06 kPa (gauge)**

Example 2. A differential mercury manometer between two points of a water pipe shows 0.1 m difference. Find the pressure difference.

Solution:

1. $\Delta p = 9.81 \times 0.1 \times (13600 - 1000) =$ **12.36 kPa**

Example 3. A Bourdon gauge reads 250 kPa; atmospheric pressure is 101.3 kPa. Find the absolute pressure.

Solution:

1. $p_{abs} = 250 + 101.3 =$ **351.3 kPa**

Example 4. An inclined manometer ($\theta = 30°$) reads 40 cm of water. Find the pressure.

Solution:

1. $h = 0.4 \times \sin 30° = 0.2$ m
2. $p = 9810 \times 0.2 =$ **1962 Pa**

Example 5. Find the water column equivalent to 760 mm of mercury.

Solution:

1. $h_w = 13.6 \times 0.76 =$ **10.34 m**

6. Frequently Asked Facts

- A piezometer cannot measure gas or negative pressure.
- Bourdon tube — high pressure, gauge pressure.
- Differential manometer — pressure difference between two points; inverted U-tube — small differences.
- Absolute = atmospheric + gauge.

7. Note (Confusing Facts)

- A barometer measures atmospheric pressure, not gauge pressure.
- Under vacuum, absolute = atmospheric − vacuum.
- The dead-weight tester is for calibration, not routine measurement.

Remember: One line: going down add ρgh, going up subtract ρgh. Assertion–Reason: (A) A mercury manometer suits high pressures. (R) Mercury has high density. — Both true; R explains A.

30 Most Important One-Liner Facts — Measurement of Fluid Pressure

1. Absolute pressure?

— Atmospheric + gauge.

2. Pressure below atmospheric?

— Vacuum pressure.

3. Instrument for atmospheric pressure?

— Barometer.

4. Simplest manometer?

— Piezometer.

5. Limitation of piezometer?

— No gas or negative pressure.

6. Pressure in simple U-tube?

— $\rho_2 g h_2 - \rho_1 g h_1$.

7. Pressure difference between two points?

— Differential manometer.

8. Small pressure difference?

— Inverted U-tube / micromanometer.

9. h in inclined manometer?

— L sin θ.

10. Mechanical gauge for high pressure?

— Bourdon tube.

11. Section of Bourdon tube?

— Elliptical.

12. Bourdon gauge reads?

— Gauge pressure.

13. Mechanical gauge for low pressure?

— Diaphragm gauge.

14. Calibration of gauges?

— Dead-weight tester.

15. Common heavy manometric liquid?

— Mercury.

16. Specific gravity of mercury?

— 13.6.

17. 760 mm Hg = ? m water

— 10.34.

18. Pressure head?

— p/ρg.

19. Velocity from Pitot tube?

— $\sqrt{(2gh)}$.

20. Same liquid, same level pressure?

— Equal.

21. Going down in manometer?

— $+\rho gh$.

22. Going up in manometer?

— $-\rho gh$.

23. Aneroid barometer?

— Barometer without liquid.

24. Differential U-tube formula?

— $gh(\rho_g - \rho)$.

25. Vacuum gauge shows?

— Amount below atmospheric.

26. Benefit of inclined manometer?

— Higher sensitivity.

27. Liquid in inverted U-tube?

— Lighter (air/oil).

28. Property of manometric liquid?

— Immiscible, low vapour pressure.

29. Atmospheric pressure at sea level?

— 101.325 kPa.

30. Bellows gauge?

— Expansion of elastic bellows.

Practice MCQs — Measurement of Fluid Pressure

Q1. A piezometer cannot measure:

(a) Low positive pressure
(b) Negative pressure
(c) Liquid pressure
(d) Water pressure

✔ **Answer: (b) Negative pressure**

Explanation: Liquid would be sucked.

Q2. A Bourdon tube gauge measures:

(a) Absolute pressure
(b) Gauge pressure
(c) Atmospheric pressure
(d) Velocity

✔ **Answer: (b) Gauge pressure**

Explanation: Relative to atmosphere.

Q3. Absolute pressure =

(a) Gauge − atmospheric
(b) Gauge + atmospheric
(c) Gauge only
(d) Atmospheric − gauge

✔ **Answer: (b) Gauge + atmospheric**

Explanation: Definition.

Q4. Pressure difference between two pipes is measured by:

(a) Piezometer
(b) Differential manometer
(c) Barometer
(d) Pitot tube

✔ **Answer: (b) Differential manometer**

Explanation: Differential.

Q5. Inclined manometer (30°), reading 40 cm — vertical height:

(a) 40 cm
(b) 20 cm
(c) 34.6 cm
(d) 80 cm

✔ **Answer: (b) 20 cm**

Explanation: L sin 30°.

Q6. Cross-section of a Bourdon tube:

(a) Circular
(b) Elliptical
(c) Square
(d) Triangular

✔ **Answer: (b) Elliptical**

Explanation: Tends to become round under pressure.

Q7. Mercury is used in manometers because of its:

(a) Low density
(b) High density and low vapour pressure
(c) Colour
(d) Cheapness

✔ **Answer: (b) High density and low vapour pressure**

Explanation: Short column.

Q8. Bourdon gauge 200 kPa, atmosphere 100 kPa — absolute:

(a) 100 kPa
(b) 200 kPa
(c) 300 kPa
(d) 2 kPa

✔ **Answer: (c) 300 kPa**

Explanation: 200 + 100.

Q9. Used for calibrating pressure gauges:

(a) Dead-weight tester
(b) Piezometer
(c) Venturimeter
(d) Barometer

✔ **Answer: (a) Dead-weight tester**

Explanation: Standard weights.

Q10. Differential manometer (mercury–water), h = 0.1 m — Δp:

(a) 13.34 kPa
(b) 12.36 kPa
(c) 9.81 kPa
(d) 1.36 kPa

✔ **Answer: (b) 12.36 kPa**

Explanation: 9.81 × 0.1 × 12600.

Q11. For very small pressure differences:

(a) Bourdon gauge
(b) Inverted U-tube/micromanometer
(c) Piezometer
(d) Barometer

✔ **Answer: (b) Inverted U-tube/micromanometer**

Explanation: Lighter liquid.

Q12. Atmospheric pressure is measured by:

(a) Manometer
(b) Barometer
(c) Bourdon gauge
(d) Pitot tube

✔ **Answer: (b) Barometer**

Explanation: Barometer.

Q13. Pressure head =

(a) $p \times \rho g$
(b) $p/(\rho g)$
(c) $\rho g/p$
(d) $p\rho$

✔ **Answer: (b) $p/(\rho g)$**

Explanation: $h = p/\rho g$.

Q14. Moving up in a manometer column, pressure:

(a) Increases
(b) Decreases
(c) Constant
(d) Zero

✔ **Answer: (b) Decreases**

Explanation: $-\rho gh$.

Q15. Vacuum 30 kPa, atmosphere 100 kPa — absolute:

(a) 130 kPa
(b) 70 kPa
(c) 30 kPa
(d) 100 kPa

✔ **Answer: (b) 70 kPa**

Explanation: 100 − 30.

SSC JE / UPSSSC JE Previous Year Questions — Measurement of Fluid Pressure

Q1. Not suitable for measuring negative pressure:

SSC JE (Previous Year)

(a) U-tube manometer
(b) Piezometer
(c) Differential manometer
(d) Micromanometer

✔ **Answer: (b) Piezometer**

Explanation: Piezometer.

Q2. A Bourdon pressure gauge measures:

SSC JE (Previous Year)

(a) Gauge pressure
(b) Absolute pressure
(c) Atmospheric pressure
(d) Vacuum only

✔ **Answer: (a) Gauge pressure**

Explanation: Gauge.

Q3. Absolute pressure is measured relative to: SSC JE (Previous Year)

(a) Atmospheric pressure
(b) Perfect vacuum
(c) Gauge pressure
(d) Sea level

✔ **Answer: (b) Perfect vacuum**

Explanation: Zero pressure.

Q4. An inverted U-tube differential manometer is used for: SSC JE (Previous Year)

(a) High pressure difference
(b) Low pressure difference
(c) Atmospheric pressure
(d) Velocity

✔ **Answer: (b) Low pressure difference**

Explanation: Light liquid.

Q5. 1 atmosphere is approximately: SSC JE (Previous Year)

(a) 10.3 m of water
(b) 76 m of water
(c) 1 m of mercury
(d) 100 m of air

✔ **Answer: (a) 10.3 m of water**

Explanation: 10.33 m.

UPSSSC ALERT

Chapter 3: Hydraulic Turbines

Hydraulic Turbines – Basic Concept

A **hydraulic turbine** converts the energy of water (head — potential and pressure energy) into rotary mechanical energy; driving a generator, it produces hydro-electric power. Indian plants such as Bhakra, Tehri and Rihand are based on it.

Understand in Detail — Hydraulic Turbines

3.1 Hydraulic turbines — from water's energy to electricity

Water stored at a height behind a dam has **potential energy**. As it comes down through the penstock (a large pipe), this energy turns into pressure and kinetic energy. The turbine converts it into rotation of its runner — **mechanical energy** — and the generator coupled to the runner produces electricity.

Gross head (H_g) = difference between head-race (upper reservoir) and tail-race (lower outlet) levels. **Net head (H)** = gross head − friction loss in the penstock ($h_f = 4fLV^2/2gD$). The turbine actually receives only the net head. **Water power** = ρ g Q H (W).

Efficiencies: hydraulic efficiency η_h = power given to runner ÷ water power; **mechanical efficiency** η_m = shaft power ÷ runner power; **volumetric efficiency** η_v = water actually striking runner ÷ water supplied; **overall efficiency η_o = shaft power/(ρgQH) = $\eta_h \times \eta_m$** (times η_v).

Classification: (1) **by type of energy** — **impulse**: the whole pressure head is converted to velocity in the nozzle before reaching the runner, which works at atmospheric pressure (Pelton); **reaction**: on entering the runner the water has both pressure and velocity, both change in the runner, and the runner runs full of water in a closed casing (Francis, Kaplan). (2) **by flow direction** — tangential (Pelton), radial/mixed (Francis), axial (Kaplan, propeller). (3) **by head** — high (> 250 m) Pelton, medium (60–250 m) Francis, low (< 60 m) Kaplan.

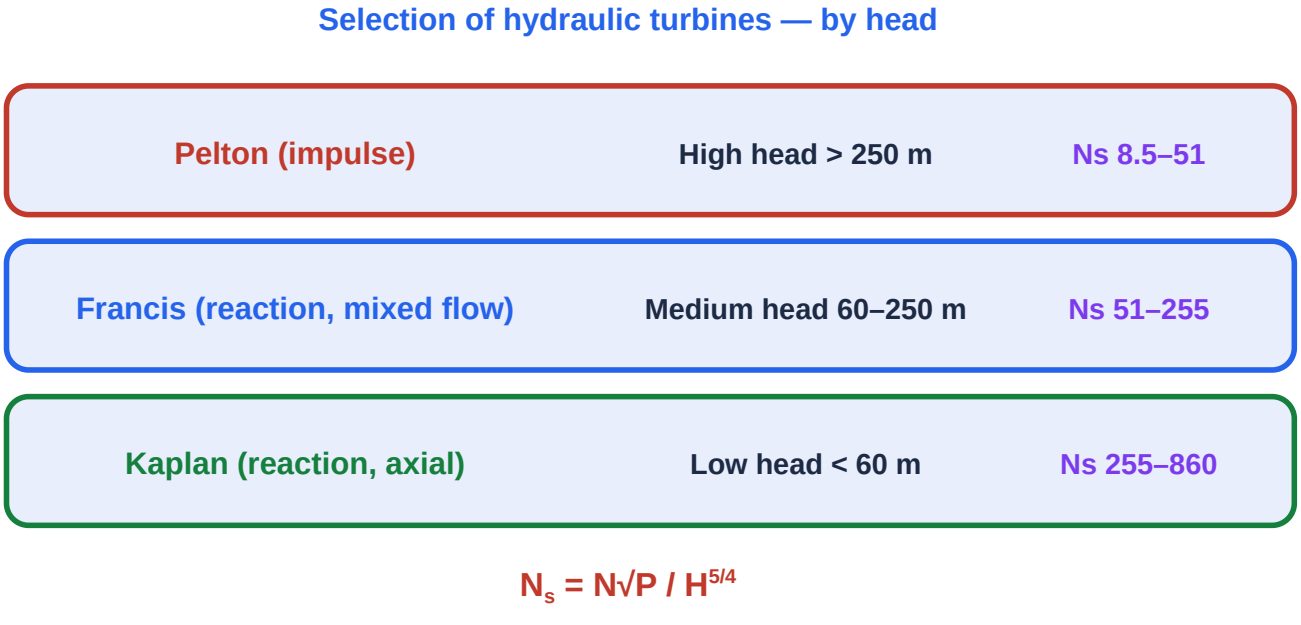


Fig 8.3: Selection of turbines by head and specific speed

3.2 Pelton wheel — understanding the impulse turbine in depth

Construction: (1) **Nozzle and spear (needle):** at the end of the penstock the nozzle converts the whole pressure into a fast jet; a conical spear inside moves to and fro to vary the jet area (hence Q) — control according to load. (2) **Runner and buckets:** double hemispherical cup-shaped buckets on the wheel rim, with a sharp edge between them — the **splitter** — dividing the jet into two equal parts so that the axial thrust is balanced. (3) **Casing:** only to prevent splashing, it has no hydraulic function. (4) **Braking jet:** when shutting down, a jet on the back of the buckets stops the wheel.

How is work done? The jet strikes the bucket and is turned through about 165° (not 180°, so that the leaving water does not hit the next bucket). The change in direction changes the momentum of water, and by Newton's second law this change becomes the force on the bucket.

Derivation: Force on bucket, work and maximum efficiency

1. Jet velocity $V = C_v \sqrt{(2gH)}$ ($C_v \approx 0.97–0.99$) ; bucket velocity $u = \pi DN/60$
2. Velocity relative to bucket at inlet = $V - u$; at outlet (friction neglected) same magnitude, turned through deflection angle φ
3. Force in the direction of motion $F = \rho Q (V - u)(1 + \cos \beta)$, where $\beta = 180° - \varphi$ (outlet angle)
4. Work per second = $F u = \rho Q (V - u)(1 + \cos \beta) u$; hydraulic efficiency $\eta_h = 2(V - u)(1 + \cos \beta) u / V^2$
5. $d\eta/du = 0 \rightarrow V - 2u = 0 \rightarrow$ efficiency is maximum at **$u = V/2$**
6. **$\eta_{max} = (1 + \cos \beta)/2$** ; if $\beta = 0$ (full 180° turn) it is 100% — in practice $\beta \approx 15°$, $\eta_{max} \approx 98\%$

In practice the bucket speed is kept $u \approx 0.46\ V$ (**speed ratio $\varphi = u/\sqrt{(2gH)} \approx 0.43–0.47$**). The **jet ratio $m = D/d$** (wheel diameter ÷ jet diameter) is usually around 12; number of buckets ≈ 15 + D/(2d).

Example: A Pelton wheel works under a net head of 400 m, $C_v = 0.98$. Find the jet velocity and the bucket velocity for maximum efficiency.

Solution:

1. $V = 0.98 \times \sqrt{(2 \times 9.81 \times 400)} = 0.98 \times 88.59 =$ **86.8 m/s**
2. $u = V/2 =$ **43.4 m/s** ; if N = 500 rpm then $D = 60u/(\pi N) = 60 \times 43.4/(\pi \times 500) =$ **1.66 m**

3.3 Francis and Kaplan — reaction turbines

Parts of a Francis turbine (in order of water path): (1) **scroll/spiral casing** — snail-shaped, with gradually decreasing area so that water is distributed with uniform velocity all round; (2) **stay vanes** — fixed, support the structure; (3) **guide vanes (wicket gates)** — can be rotated, turn water at the correct angle towards the runner and control the flow (connected to the governor); (4) **runner** — where water enters radially and leaves axially (so a modern Francis = **mixed flow**); (5) **draft tube** — outlet.

Kaplan turbine: for very low head and very large flow (barrages on rivers). The runner is like a ship's propeller — a large hub (boss) with only 3–8 blades; water flows fully **axially**. Its biggest feature: **the runner blade angle can also be changed while running** (by a servo-motor inside the hub). So efficiency remains high even at part load. If the blades are fixed it is called a **propeller turbine**.

Draft tube — why needed? It is a gradually widening (diverging) tube from the runner outlet to the tail-race, with its lower end submerged in tail-race water. Two jobs: (1) even when the runner is placed above the tail-race, that height of head is not wasted — **negative pressure (suction)** forms at the runner outlet; (2) in the widening tube the velocity of leaving water falls, so **most of its kinetic energy is recovered as pressure**. A draft tube is used only with reaction turbines, not with a Pelton.

Degree of reaction (R) = change of pressure energy in the runner ÷ total energy change in the runner. For a Pelton R = 0; for Francis/Kaplan $0 < R < 1$.

3.4 Specific speed, unit quantities, cavitation and governing

Specific speed (N_s) — the speed of an imaginary, geometrically similar turbine that would develop **1 kW of power under 1 m head**. It is the identity of a turbine's 'type' — knowing H, P and N for a site, we can find N_s and immediately tell which turbine to install.

Derivation: Formula for N_s (in brief)

1. For similar turbines $u \propto \sqrt{H}$ and $u = \pi DN/60 \rightarrow N \propto \sqrt{H}/D$
2. $Q \propto$ area × velocity $\propto D^2 \sqrt{H}$; $P \propto Q\,H \propto D^2 H^{3/2} \rightarrow D \propto \sqrt{P} / H^{3/4}$
3. $N \propto \sqrt{H} \times H^{3/4}/\sqrt{P} \rightarrow N\sqrt{P} / H^{5/4} =$ constant
4. $\mathbf{N_s = N\sqrt{P} / H^{5/4}}$ (P in kW)

Selection by N_s: Pelton (single jet) 8.5–30, (multi-jet) 30–60; Francis 50–340; Kaplan/propeller 255–860. **Read the formula:** high head (large H) → small N_s → Pelton; low head → large N_s → Kaplan.

Unit quantities — predicting the behaviour of the same turbine under another head: unit speed $N_u = N/\sqrt{H}$, unit discharge $Q_u = Q/\sqrt{H}$, unit power $P_u = P/H^{3/2}$. That is, when the head changes **$N \propto \sqrt{H}$, $Q \propto \sqrt{H}$, $P \propto H^{3/2}$**.

Cavitation: at the runner outlet or top of the draft tube, where the pressure is lowest, if the pressure falls below the vapour pressure of water, the water 'boils' right there and vapour bubbles form. As these bubbles are carried into a high-pressure region they suddenly **collapse** — a micro-shock of thousands of bar hits the metal surface, causing pitting, noise, vibration and loss of efficiency. **Prevention:** setting the turbine close to or below tail-race level (limited setting height), keeping the Thoma cavitation factor σ above its critical value, using hard metal like stainless steel.

Governing: controlling the water flow so that the turbine speed (and hence the 50 Hz frequency) stays constant even as the electrical load varies. Load falls → speed rises → governor fly-balls rise → oil-pressure servo-motor moves the spear forward in a Pelton (or closes the guide vanes in a Francis) → flow reduces → speed returns to normal. In a Pelton, suddenly closing the spear could cause **water hammer** in the penstock, so first a **deflector** moves the jet away from the buckets and the spear closes slowly.

> **Example:** At a site H = 25 m, P = 5000 kW, N = 250 rpm. Which turbine is suitable?
>
> **Solution:**
>
> 1. $\sqrt{P} = \sqrt{5000} = 70.7$; $H^{5/4} = 25^{1.25} = 55.9$
> 2. $N_s = 250 \times 70.7/55.9 =$ **316**
> 3. 316 — at the upper limit of Francis and the lower limit of Kaplan; considering the low head (25 m) a **Kaplan** is more suitable (a fast Francis is also possible)

1. Key Points

- **Impulse turbine:** all pressure energy converted to kinetic energy in the nozzle; runner at atmospheric pressure; example — **Pelton wheel** (high head, low discharge).
- **Reaction turbine:** water enters the runner with both pressure and kinetic energy; runner full of water, closed casing, draft tube; examples — **Francis** (medium head, mixed flow), **Kaplan** and propeller (low head, axial flow, high discharge).
- **Kaplan vs propeller:** Kaplan runner blades are adjustable — good efficiency even at part load; propeller blades are fixed.
- **Specific speed:** $N_s = N\sqrt{P}/H^{5/4}$ — the speed at which a geometrically similar turbine would develop 1 kW under 1 m head. It is the basis of turbine selection.
- **N_s ranges:** Pelton 8.5–30 (single jet), 30–51 (multi-jet); Francis 51–255; Kaplan 255–860.
- **Efficiencies:** hydraulic η_h = power developed by runner ÷ water power (ρgQH); mechanical η_m = shaft power ÷ runner power; overall $\eta_o = \eta_h \times \eta_m$ = shaft power/ρgQH.
- **Pelton wheel:** jet velocity $V = C_v\sqrt{(2gH)}$ ($C_v \approx 0.98$); for maximum efficiency bucket speed $u = V/2$; jet deflection on bucket ≈ 165°; jet ratio $m = D/d$ (≈ 12).

- **Draft tube:** a gradually expanding tube from the reaction turbine outlet to the tail race; it allows the turbine to be placed above tail race and recovers part of the exit kinetic energy (negative pressure at runner exit).
- **Governing:** Pelton by spear (needle) and deflector; Francis by guide vane (wicket gate) angle; Kaplan by both guide vanes and runner blades.
- **Ancillary parts:** penstock (large pipe from reservoir to turbine), surge tank (protection from water hammer), scroll (spiral) casing.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Water power	$P = \rho g Q H$
Overall efficiency	$\eta_o = P_{shaft} / (\rho g Q H) = \eta_h \times \eta_m$
Specific speed	$N_s = N \sqrt{P} / H^{5/4}$
Jet velocity (Pelton)	$V = C_v \sqrt{(2gH)}$
Pelton — max efficiency	$u = V/2$
Bucket speed	$u = \pi DN/60$
Jet ratio	$m = D/d$
Unit speed	$N_u = N/\sqrt{H}$
Unit discharge / power	$Q_u = Q/\sqrt{H}$, $P_u = P/H^{3/2}$

3. Detailed Explanation

- **Selecting a turbine:** high head, low discharge → Pelton; medium head and flow → Francis; low head, large discharge → Kaplan/propeller. With increasing specific speed: Pelton → Francis → Kaplan.
- **Direction of flow:** Pelton — tangential; Francis — inward radial with axial exit (mixed); Kaplan — axial.
- **Cavitation:** where local pressure falls below vapour pressure, vapour bubbles form and collapse — metal pitting, vibration, noise and loss of efficiency. Most likely at reaction turbine runner exit and draft tube inlet; checked by Thoma's cavitation factor σ.
- **Main and operating characteristics:** curves of unit power, discharge and efficiency against unit speed at constant head; operating curves at constant speed (generator) for load changes.

4. Comparison Table — Pelton, Francis and Kaplan

Basis	Pelton	Francis	Kaplan
Type	Impulse	Reaction	Reaction
Head	High (> 250 m)	Medium (60–250 m)	Low (< 60 m)
Flow	Tangential	Mixed (radial → axial)	Axial
N_s	8.5–51	51–255	255–860
Draft tube	No	Yes	Yes
Governing	Spear/deflector	Guide vanes	Guide vanes + runner blades

5. Solved Numerical Problems

Example 1. A turbine develops 850 kW shaft power under 50 m head with 2 m^3/s. Find the overall efficiency.

Solution:

1. Water power = $1000 \times 9.81 \times 2 \times 50 = 981$ kW
2. $\eta_o = 850/981 =$ **86.6%**

Example 2. A turbine develops 5000 kW at 250 rpm under 25 m head. Find specific speed and type.

Solution:

1. $N_s = 250 \times \sqrt{5000}/25^{1.25} = 250 \times 70.71/55.9$
2. $N_s =$ **316 → Kaplan**

Example 3. A Pelton wheel works under 400 m head, $C_v = 0.98$. Find jet velocity and bucket speed for maximum efficiency.

Solution:

1. $V = 0.98 \times \sqrt{(2 \times 9.81 \times 400)} = 0.98 \times 88.59 =$ **86.8 m/s**
2. $u = V/2 =$ **43.4 m/s**

Example 4. If the bucket circle diameter above is 1.5 m, find the speed (rpm).

Solution:

1. $N = 60u/(\pi D) = 60 \times 43.4/(\pi \times 1.5) =$ **552.6 rpm**

Example 5. Water power 1000 kW, $\eta_h = 0.9$, $\eta_m = 0.95$. Find the shaft power.

Solution:

1. $\eta_o = 0.9 \times 0.95 = 0.855$
2. $P = 0.855 \times 1000 =$ **855 kW**

6. Frequently Asked Facts

- Pelton — high head, impulse; Francis — medium head; Kaplan — low head.

- $N_s = N\sqrt{P}/H^{5/4}$ (turbine).
- Pelton maximum efficiency: u = V/2.
- Draft tube only in reaction turbines.

7. Note (Confusing Facts)

- Kaplan is axial, Francis is mixed — calling Francis axial is wrong.
- Turbine N_s has $\sqrt{P}$; pump N_s has $\sqrt{Q}$ and $H^{3/4}$.
- The Pelton runner is at atmospheric pressure — hence no draft tube.

Remember: One line: head ↓, N_s ↑: Pelton → Francis → Kaplan. Assertion–Reason: (A) A Kaplan turbine stays efficient at part load. (R) Its runner blades are adjustable. — Both true; R explains A.

30 Most Important One-Liner Facts — Hydraulic Turbines

1. Function of hydraulic turbine?

— Water energy → mechanical energy.

2. Impulse hydraulic turbine?

— Pelton wheel.

3. For high head?

— Pelton.

4. For medium head?

— Francis.

5. For low head?

— Kaplan.

6. Mixed-flow turbine?

— Francis.

7. Axial-flow turbine?

— Kaplan / propeller.

8. Adjustable runner blades?

— Kaplan.

9. Specific speed (turbine)?

— $N\sqrt{P}/H^{5/4}$.

10. N_s of Pelton?

— 8.5–51.

11. N_s of Francis?

— 51–255.

12. N_s of Kaplan?

— 255–860.

13. Water power?

— ρgQH.

14. Overall efficiency?

— $\eta_h \times \eta_m$.

15. Jet velocity?

— $C_v\sqrt{(2gH)}$.

16. Pelton max efficiency condition?

— u = V/2.

17. Jet ratio?

— D/d (≈ 12).

18. Deflection on Pelton bucket?

— ≈ 165°.

19. Draft tube in?

— Reaction turbines.

20. Function of draft tube?

— Recover KE, place turbine above tail race.

21. Pelton governing?

— Spear and deflector.

22. Francis governing?

— Guide vanes.

23. Pipe from reservoir to turbine?

— Penstock.

24. Protection from water hammer?

— Surge tank.

25. Pressure below vapour pressure?

— Cavitation.

26. Cavitation factor?

— Thoma's σ.

27. Effect of cavitation?

— Pitting, vibration, loss of efficiency.

28. Unit speed?

— $N/\sqrt{H}$.

29. Flow direction in Pelton?

— Tangential.

30. Pressure on Pelton runner?

— Atmospheric.

Practice MCQs — Hydraulic Turbines

Q1. Pelton wheel is a:

(a) Reaction turbine
(b) Impulse turbine
(c) Axial flow
(d) Mixed flow

✔ **Answer: (b) Impulse turbine**

Explanation: Tangential impulse.

Q2. Suitable for low head and large discharge:

(a) Pelton
(b) Francis
(c) Kaplan
(d) Turgo

✔ **Answer: (c) Kaplan**

Explanation: Axial.

Q3. Specific speed of a turbine:

(a) $N\sqrt{Q}/H^{3/4}$
(b) $N\sqrt{P}/H^{5/4}$
(c) $N\sqrt{P}/H^{3/4}$
(d) NP/H

✔ **Answer: (b) $N\sqrt{P}/H^{5/4}$**

Explanation: Turbine.

Q4. Bucket speed for max efficiency of a Pelton wheel:

(a) V
(b) V/2
(c) 2V
(d) V/4

✔ **Answer: (b) V/2**

Explanation: u = V/2.

Q5. A draft tube is used in:

(a) Pelton
(b) Reaction turbines
(c) Centrifugal pumps
(d) Reciprocating pumps

✔ **Answer: (b) Reaction turbines**

Explanation: Francis, Kaplan.

Q6. Francis turbine is a:

(a) Axial flow impulse
(b) Mixed flow reaction
(c) Tangential impulse
(d) Outward radial impulse

✔ **Answer: (b) Mixed flow reaction**

Explanation: Mixed.

Q7. Turbine with adjustable runner blades:

(a) Propeller
(b) Kaplan
(c) Francis
(d) Pelton

✔ **Answer: (b) Kaplan**

Explanation: Kaplan.

Q8. Water power =

(a) ρgQ
(b) ρgQH
(c) ρQH
(d) gQH/ρ

✔ **Answer: (b) ρgQH**

Explanation: W.

Q9. Cavitation occurs when pressure falls below:

(a) Atmospheric
(b) Vapour pressure
(c) Gauge pressure
(d) Critical pressure

✔ **Answer: (b) Vapour pressure**

Explanation: Vapour bubbles.

Q10. N = 250 rpm, P = 5000 kW, H = 25 m — N_s about:

(a) 32
(b) 160
(c) 316
(d) 700

✔ **Answer: (c) 316**

Explanation: 250 × 70.7/55.9.

Q11. Governing of a Pelton wheel is by:

(a) Guide vanes
(b) Spear (needle) and deflector
(c) Runner blades
(d) Draft tube

✔ **Answer: (b) Spear (needle) and deflector**

Explanation: Jet control.

Q12. To protect the penstock from water hammer:

(a) Draft tube
(b) Surge tank
(c) Scroll casing
(d) Nozzle

✔ **Answer: (b) Surge tank**

Explanation: Surge tank.

Q13. H = 400 m, C_v = 0.98 — jet velocity about:

(a) 44 m/s
(b) 87 m/s
(c) 120 m/s
(d) 196 m/s

✔ **Answer: (b) 87 m/s**

Explanation: 0.98 × 88.6.

Q14. Correct order of increasing specific speed:

(a) Kaplan, Francis, Pelton
(b) Pelton, Francis, Kaplan
(c) Francis, Pelton, Kaplan
(d) Pelton, Kaplan, Francis

✔ **Answer: (b) Pelton, Francis, Kaplan**

Explanation: Low → high.

Q15. η_h = 0.9, η_m = 0.95 — η_o:

(a) 0.95
(b) 0.855
(c) 0.9
(d) 1.85

✔ **Answer: (b) 0.855**

Explanation: Product.

SSC JE / UPSSSC JE Previous Year Questions — Hydraulic Turbines

Q1. A Pelton wheel is suitable for:

SSC JE (Previous Year)

(a) Low head
(b) Medium head
(c) High head
(d) Any head

✔ **Answer: (c) High head**

Explanation: High head, low flow.

Q2. Kaplan turbine is a:

SSC JE (Previous Year)

(a) High head mixed flow
(b) Low head axial flow
(c) Impulse
(d) Tangential

✔ **Answer: (b) Low head axial flow**

Explanation: Axial.

Q3. Purpose of the draft tube in a turbine:

SSC JE (Previous Year)

(a) Increase pressure
(b) Recover exit kinetic energy
(c) Change water direction
(d) Increase cavitation

✔ **Answer: (b) Recover exit kinetic energy**

Explanation: Raises efficiency.

Q4. Water flow in a Francis turbine is: SSC JE (Previous Year)

(a) Tangential
(b) Inward radial/mixed
(c) Outward radial
(d) Axial

✔ **Answer: (b) Inward radial/mixed**

Explanation: Mixed.

Q5. Formula for specific speed of a hydraulic turbine: SSC JE (Previous Year)

(a) $N\sqrt{P}/H^{5/4}$
(b) $N\sqrt{Q}/H^{3/4}$
(c) $N\sqrt{P}/H^{3/2}$
(d) NQ/H

✔ **Answer: (a) $N\sqrt{P}/H^{5/4}$**

Explanation: Turbine.

Q6. Pelton buckets deflect the jet through about: SSC JE (Previous Year)

(a) 90°
(b) 120°
(c) 165°
(d) 180°

✔ **Answer: (c) 165°**

Explanation: ≈ 165°.

UPSSSC ALERT

Chapter 4: Centrifugal Pumps — Classification, Principles & Performance

Centrifugal Pumps — Classification, Principles & Performance – Basic Concept

A **pump** is a hydraulic machine that takes mechanical energy and raises the pressure energy of a liquid. In a **centrifugal pump** the rotating impeller throws the liquid outward by centrifugal action; it works as the reverse of a turbine. It is the most widely used pump — water supply, irrigation, boiler feed and industry.

Understand in Detail — Centrifugal Pumps — Classification, Principles & Performance

4.1 Centrifugal pump — understanding the principle from the root

Fill a bucket with water and whirl it fast on a rope — the water does not fall out, but presses on the bottom of the bucket. If there is a hole in the bottom, water shoots out. This is **centrifugal action**: an outward force acts on the rotating liquid and its pressure increases with radius. A centrifugal pump uses this principle to convert **mechanical energy into pressure energy of the liquid** — that is, it is a **reaction turbine running in reverse**.

Main parts: (1) **impeller** — a rotating wheel with backward-curved vanes; (2) **casing** — around the impeller; (3) **suction pipe** — with a **foot valve** at the lower end (opens only upward, so the pipe does not empty when stopped) and a **strainer**; (4) **delivery pipe** — with a **delivery valve** to regulate the flow.

Water path: water enters the **eye** (centre) of the impeller axially from the suction pipe → the rotating vanes throw it radially outward, increasing both its velocity and pressure → at the impeller exit the water velocity is very high → the casing **converts this velocity into pressure** → water at high pressure into the delivery pipe.

Types of casing: (1) **volute casing** — spiral-shaped with gradually increasing area, so velocity falls and pressure rises; simple but some eddy loss. (2) **vortex casing** — a circular chamber between impeller and volute in which a free vortex forms; less loss. (3) **diffuser (guide-blade) casing** — fixed guide vanes around the impeller lead the water through gradually widening passages; highest efficiency — such a pump is also called a **turbine pump**.

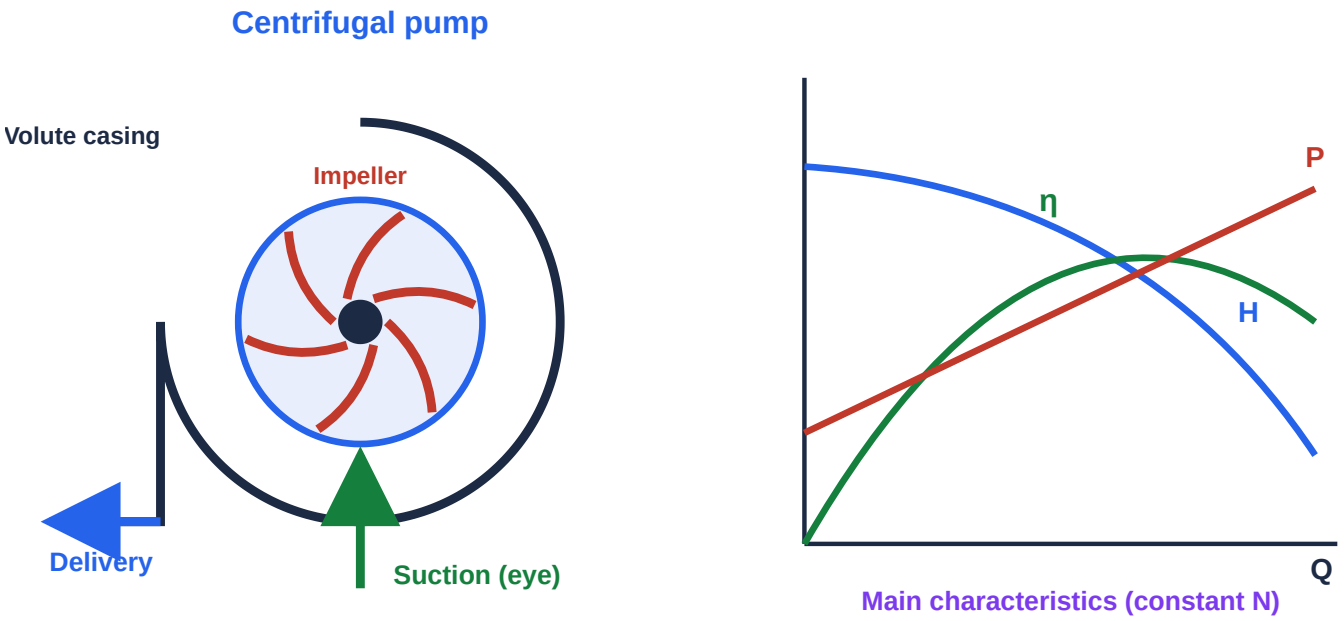


Fig 8.4: Centrifugal pump and its main characteristics

4.2 Priming — why must the pump be 'filled' with water?

The head produced by the impeller is **proportional to u^2/g** — that is, it **does not depend on the density** of the fluid, but the pressure produced by that head, $p = \rho g H$, is proportional to density. If the impeller is full of air, it will produce the same 'metres' of head but in **metres of air**! Air is about 800 times lighter than water, so the pressure produced is so small that it cannot lift water even a few centimetres up the suction pipe.

Priming = completely filling the suction pipe, casing and the part up to the delivery valve with water before starting, so that all air is expelled. The foot valve stops the water from falling back. Large pumps are primed by drawing out air with a **vacuum pump**; some pumps are **self-priming**.

Correct starting sequence: prime → start the motor **with the delivery valve closed** (at a closed valve the flow is zero, so the pump takes minimum power — less load on the motor) → when full speed is reached, open the delivery valve slowly. While stopping too, close the delivery valve first and then the motor — so that water does not flow back.

> **Note:** In a centrifugal pump the power is minimum at closed delivery; in contrast a **reciprocating pump** must never be run with the delivery valve closed — the pressure would rise indefinitely and the pipe could burst, so it must have a **relief valve**.

4.3 Heads, work and efficiencies — with the help of velocity triangles

Static head (H_s) = suction head h_s (from sump water level to pump centre) + delivery head h_d (from pump centre to water level of the upper tank). **Manometric head (H_m)** = the head against which the pump has to work = H_s + friction losses in pipes + velocity head of water leaving the delivery ($V_d^2/2g$). H_m can also be found by adding the readings of the vacuum gauge and pressure gauge at pump inlet and outlet.

Velocity triangles: at any point on the impeller the absolute velocity of water (V) = vector sum of vane velocity (u, peripheral) and velocity of water relative to vane (V_r). At inlet, ideally the water enters **radially** — i.e. with no whirl component: $V_{w1} = 0$.

Derivation: Work done by the impeller (Euler equation)

1. Work per kg of water = $(V_{w2}\, u_2 - V_{w1}\, u_1)$ — from the change of angular momentum
2. With radial entry ($V_{w1} = 0$): **work = $V_{w2}\, u_2$ per kg**, or **Euler head $H_e = V_{w2}\, u_2 / g$**
3. From the outlet triangle $V_{w2} = u_2 - V_{f2} \cot \beta_2$ (β_2 = vane outlet angle, V_{f2} = flow velocity)
4. **Manometric efficiency $\eta_{mano} = g\, H_m/(V_{w2}\, u_2)$** ; mechanical efficiency = power given to impeller ÷ shaft power ; **overall efficiency $\eta_o = \rho g Q H_m$/shaft power**

Vane shape: backward-curved ($\beta_2 < 90°$) vanes are the most used in practice — as flow increases the head falls and the motor load stays self-limiting (non-overloading); efficiency is also good. Forward-curved ($\beta_2 > 90°$) vanes give more head but the power keeps rising with flow and the motor may get overloaded.

Minimum starting speed: the pump starts delivering only when the centrifugal head developed in the impeller, $(u_2^2 - u_1^2)/2g$, equals or exceeds the manometric head. This gives the minimum speed $N_{min} = 120\, \eta_{mano}\, V_{w2}\, D_2 / [\pi(D_2^2 - D_1^2)]$.

Example: A pump delivers 0.05 m³/s of water against a manometric head of 30 m. If the overall efficiency is 75%, find the shaft power.

Solution:

1. Water power = $\rho g Q H_m = 1000 \times 9.81 \times 0.05 \times 30 = 14715$ W = 14.7 kW
2. Shaft power = 14.7/0.75 = **19.6 kW**

4.4 Affinity laws, specific speed, series-parallel and cavitation (NPSH)

Affinity laws — when the speed N of the same pump changes (same diameter): **$Q \propto N$, $H \propto N^2$, $P \propto N^3$**. Reason: flow velocity ∝ peripheral velocity ∝ N; head ∝ $u^2 \propto N^2$; power ∝ QH ∝ N^3. When the diameter D changes (same speed): $Q \propto D^3$, $H \propto D^2$, $P \propto D^5$.

Specific speed of a pump: the speed of an imaginary similar pump delivering **1 m³/s against 1 m head**: **$N_s = N \sqrt{Q} / H^{3/4}$**. Note — a turbine uses √P and $H^{5/4}$, a pump √Q and $H^{3/4}$. Low N_s → radial flow (high head, low flow); high N_s → mixed/axial flow.

Series and parallel: when a very high head is needed (e.g. a tall building) pumps are put **in series** — the same water passes through all impellers in turn: **total head = n × H, same discharge** (multistage pump). When a very large flow is needed, **in parallel — total discharge = n × Q, same head**.

Cavitation in pumps and NPSH: the lowest pressure in a pump is at the **eye (inlet)** of the impeller. If the suction lift is too high or the water is hot (higher vapour pressure), the pressure there falls below the vapour pressure and cavitation occurs — noise ('gravel-like sound'), vibration, drop in head and efficiency, pitting of vanes. **NPSH (Net Positive Suction Head)** = total head at pump inlet − vapour

pressure head = $(p_{atm}/\rho g) - h_s - h_{fs} - (p_v/\rho g)$. Rule: **NPSH available > NPSH required** (given by the maker). That is why the suction lift is theoretically below 10.33 m and in practice kept only up to about 6–7 m.

Example: A pump runs at 1450 rpm giving 0.03 m³/s at 20 m head with 8 kW power. What are these at 2900 rpm?

Solution:

1. Speed ratio = 2900/1450 = 2
2. Q = 0.03 × 2 = **0.06 m³/s** ; H = 20 × 2² = **80 m** ; P = 8 × 2³ = **64 kW**
3. N_s (at 1450 rpm) = 1450 × √0.03/$20^{0.75}$ = 1450 × 0.1732/9.457 = **26.6** → radial-flow pump

Note: Reciprocating vs centrifugal pump: reciprocating — small flow, very high pressure, pulsating flow, no priming needed, flow smoothed by an **air vessel**; centrifugal — large flow, moderate head, smooth flow, priming needed, compact and cheap. In a reciprocating pump **slip = $(Q_{th} - Q_{act})/Q_{th}$**; negative slip occurs when the suction pipe is long, the delivery pipe short and at high speed the delivery valve opens early.

1. Key Points

- **Main parts:** impeller (with vanes), casing (volute/spiral, vortex chamber, diffuser), suction pipe (with foot valve and strainer), delivery pipe (with delivery valve).
- **Classification:** casing — volute, vortex, diffuser; impeller — closed (shrouded), semi-open, open; flow — radial, mixed, axial; suction — single/double; stages — single/multi-stage (in series for high head).
- **Working principle:** forced vortex — rotation raises pressure head with the square of radius; velocity head is converted into pressure in the casing.
- **Priming:** filling the suction pipe, casing and impeller with liquid before starting — otherwise the negligible head developed in air will not lift water.
- **Heads:** static head H_s = suction + delivery head; manometric head $H_m = H_s$ + pipe losses + $V_d^2/2g$; Euler head = $V_{w2}u_2/g$ (radial entry).
- **Efficiencies:** manometric $\eta_{mano} = gH_m/(V_{w2}u_2)$; mechanical; overall $\eta_o = \rho gQH_m$/shaft power.
- **Specific speed (pump):** $N_s = N\sqrt{Q}/H^{3/4}$ — the speed at which a similar pump delivers 1 m³/s against 1 m head.
- **Affinity laws:** for the same pump $Q \propto N$, $H \propto N^2$, $P \propto N^3$; for similar pumps $Q \propto ND^3$, $H \propto N^2D^2$, $P \propto N^3D^5$.
- **Combination:** in series (multi-stage on one shaft) — heads add, discharge same; in parallel — discharges add, head same.
- **NPSH and cavitation:** available net positive suction head must exceed the required value, otherwise cavitation occurs at the impeller eye; suction lift is usually limited to 6–7 m.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Euler head (radial entry)	$H_e = V_{w2}\, u_2 / g$
Manometric efficiency	$\eta_{mano} = g\, H_m / (V_{w2}\, u_2)$
Overall efficiency	$\eta_o = \rho\, g\, Q\, H_m / P_{shaft}$
Specific speed (pump)	$N_s = N \sqrt{Q} / H^{3/4}$
Peripheral velocity	$u = \pi D N / 60$
Affinity laws	$Q \propto N$, $H \propto N^2$, $P \propto N^3$
Minimum starting speed	$N_{min} = 120\, \eta_{mano}\, V_{w2}\, D_2 / [\pi(D_2^2 - D_1^2)]$
Series / parallel	$H = H_1 + H_2$, $Q = Q_1 + Q_2$
Water power (output)	$P = \rho\, g\, Q\, H_m$

3. Detailed Explanation

- **Velocity triangles:** radial entry ($V_{w1} = 0$) is usually assumed; at exit $V_{w2} = u_2 - V_{f2} \cot \varphi$. Vane exit angle φ — backward-curved vanes ($\varphi < 90°$) are most common because of stable characteristics and no overloading.
- **Characteristic curves:** main (H, P, η vs Q at constant N), operating, and iso-efficiency (Muschel) curves. With backward vanes H falls as Q rises.
- **Centrifugal vs reciprocating pump:** centrifugal — large flow, low-medium head, smooth flow, needs priming; reciprocating — small flow, high head, pulsating, positive displacement, needs air vessel, has slip.
- **Starting procedure:** start with the delivery valve closed (minimum power at zero flow), then open gradually.
- **Submersible and jet pumps:** submersible — motor-pump unit immersed, multi-stage; for deep borewells. Jet pump — suction by nozzle and venturi.

4. Comparison Table — Centrifugal vs Reciprocating Pump

Basis	Centrifugal	Reciprocating
Discharge	Large	Small
Head	Low-medium	High
Flow nature	Continuous, smooth	Pulsating
Priming	Needed	Not needed (self-priming)
Viscous liquids	Suitable	Less suitable
Cost/space	Less	More

5. Solved Numerical Problems

Example 1. A pump delivers 0.05 m³/s against 30 m manometric head; shaft power 20 kW. Find the overall efficiency.

Solution:

1. $P_{water} = 1000 \times 9.81 \times 0.05 \times 30 = 14.72$ kW
2. $\eta_o = 14.72/20 =$ **73.6%**

Example 2. A pump at 1450 rpm gives 0.02 m³/s at 25 m head. Find the specific speed.

Solution:

1. $N_s = 1450 \times \sqrt{0.02}/25^{0.75} = 1450 \times 0.1414/11.18$
2. $N_s =$ **18.3**

Example 3. If the above pump speed is raised to 2900 rpm, find new discharge, head and power ratio.

Solution:

1. $Q = 0.02 \times 2 =$ **0.04 m³/s**
2. $H = 25 \times 4 =$ **100 m**
3. Power $= 2^3 =$ **8 times**

Example 4. Impeller diameter 0.3 m, speed 1500 rpm, $V_{w2} = 20$ m/s, $H_m = 40$ m. Find the manometric efficiency.

Solution:

1. $u_2 = \pi \times 0.3 \times 1500/60 = 23.56$ m/s
2. $\eta_{mano} = 9.81 \times 40/(20 \times 23.56) =$ **83.3%**

Example 5. Two identical pumps (each 30 m, 0.1 m³/s) are joined in series and in parallel. State the results.

Solution:

1. Series: head **60 m**, flow 0.1 m³/s
2. Parallel: head 30 m, flow **0.2 m³/s**

6. Frequently Asked Facts

- Pump $N_s = N\sqrt{Q}/H^{3/4}$; turbine $N_s = N\sqrt{P}/H^{5/4}$.
- $Q \propto N$, $H \propto N^2$, $P \propto N^3$.
- Priming — necessary for centrifugal pumps.
- Heads add in series; discharges add in parallel.

7. Note (Confusing Facts)

- A centrifugal pump is started with the delivery valve closed; a reciprocating pump is not.
- Backward-curved vanes are most used — not forward-curved.

- Cavitation in a pump is most at the impeller eye; in a turbine at the runner exit.

Remember: One line: pump — reverse of turbine; mechanical energy → pressure energy. Assertion–Reason: (A) A centrifugal pump needs priming. (R) The pressure developed by the impeller in air is negligible. — Both true; R explains A.

30 Most Important One-Liner Facts — Centrifugal Pumps — Classification, Principles & Performance

1. Function of pump?

— Mechanical → pressure energy.

2. Rotating part of centrifugal pump?

— Impeller.

3. Spiral casing?

— Volute.

4. Working principle?

— Forced vortex.

5. Filling with water before start?

— Priming.

6. At the bottom of suction pipe?

— Foot valve and strainer.

7. Most used vanes?

— Backward-curved.

8. Euler head?

— $V_{w2}u_2/g$.

9. Manometric efficiency?

— $gH_m/(V_{w2}u_2)$.

10. Overall efficiency?

— ρgQH_m/shaft power.

11. Pump specific speed?

— $N\sqrt{Q}/H^{3/4}$.

12. Relation of Q and N?

— $Q \propto N$.

13. Relation of H and N?

— $H \propto N^2$.

14. Relation of P and N?

— $P \propto N^3$.

15. Pump arrangement for high head?

— Series (multi-stage).

16. Arrangement for more flow?

— Parallel.

17. NPSH stands for?

— Net Positive Suction Head.

18. Where is cavitation in pump?

— Impeller eye.

19. Usual max suction lift?

— 6–7 m.

20. Delivery valve at start?

— Closed.

21. Static head?

— Suction + delivery head.

22. Extra in manometric head?

— Losses + delivery velocity head.

23. Pump for deep borewell?

— Submersible.

24. Self-priming pump?

— Reciprocating.

25. Pump with pulsating flow?

— Reciprocating.

26. To reduce pulsation in reciprocating?

— Air vessel.

27. Usual inlet assumption?

— Radial entry ($V_{w1} = 0$).

28. Q in similar pumps?

— $\propto ND^3$.

29. H in similar pumps?

— $\propto N^2D^2$.

30. Speed doubled — power?

— 8 times.

Practice MCQs — Centrifugal Pumps — Classification, Principles & Performance

Q1. Priming is necessary in a centrifugal pump because:

(a) Motor stays cool

(b) Pressure developed in air is negligible

(c) Speed increases

(d) Efficiency falls

✔ **Answer: (b) Pressure developed in air is negligible**

Explanation: Air density is very low.

Q2. Specific speed of a pump:

(a) $N\sqrt{P}/H^{5/4}$

(b) $N\sqrt{Q}/H^{3/4}$

(c) $N\sqrt{Q}/H^{5/4}$

(d) NQ/H

✔ **Answer: (b) $N\sqrt{Q}/H^{3/4}$**

Explanation: Pump.

Q3. Doubling the pump speed, the head becomes:

(a) Double

(b) Four times

(c) Eight times

(d) Half

✔ **Answer: (b) Four times**

Explanation: $H \propto N^2$.

Q4. Doubling the pump speed, power becomes:

(a) 2 times

(b) 4 times

(c) 8 times

(d) 16 times

✔ **Answer: (c) 8 times**

Explanation: $P \propto N^3$.

Q5. For high head, pumps are connected in:

(a) Parallel

(b) Series

(c) Any

(d) Reverse

✔ **Answer: (b) Series**

Explanation: Heads add.

Q6. Most commonly used vanes in centrifugal pumps:

(a) Forward-curved

(b) Radial

(c) Backward-curved

(d) Straight

✔ **Answer: (c) Backward-curved**

Explanation: Stable characteristics.

Q7. Delivery valve while starting a centrifugal pump:

(a) Fully open
(b) Closed
(c) Half open
(d) Removed

✔ **Answer: (b) Closed**

Explanation: Minimum power.

Q8. Q = 0.05 m^3/s, H = 30 m, shaft 20 kW — η_o:

(a) 60%
(b) 73.6%
(c) 85%
(d) 92%

✔ **Answer: (b) 73.6%**

Explanation: 14.72/20.

Q9. Cavitation in a pump occurs at:

(a) Delivery
(b) Impeller eye (inlet)
(c) Casing
(d) Motor

✔ **Answer: (b) Impeller eye (inlet)**

Explanation: Lowest pressure.

Q10. Manometric efficiency =

(a) $gH_m/(V_{w2}u_2)$
(b) $V_{w2}u_2/(gH_m)$
(c) $\rho gQH/P$
(d) H_m/H_s

✔ **Answer: (a) $gH_m/(V_{w2}u_2)$**

Explanation: Manometric ÷ Euler head.

Q11. Working principle of a centrifugal pump:

(a) Free vortex
(b) Forced vortex
(c) Pascal
(d) Archimedes

✔ **Answer: (b) Forced vortex**

Explanation: Pressure by rotation.

Q12. Compared to a reciprocating pump, a centrifugal pump:

(a) Low flow, high head
(b) High flow, low head
(c) Self-priming
(d) Pulsating flow

✔ **Answer: (b) High flow, low head**

Explanation: Centrifugal.

Q13. Two identical pumps in parallel — discharge:

(a) Half
(b) Same
(c) Double
(d) Four times

✔ **Answer: (c) Double**

Explanation: Q adds.

Q14. Function of the foot valve in the suction pipe:

(a) Increase pressure
(b) Prevent water from flowing back
(c) Measure flow
(d) Reduce temperature

✔ **Answer: (b) Prevent water from flowing back**

Explanation: Keeps the pump primed.

Q15. N = 1450, Q = 0.02, H = 25 — N_s about:

(a) 9
(b) 18
(c) 36
(d) 72

✔ **Answer: (b) 18**

Explanation: 1450 × 0.1414/11.18.

SSC JE / UPSSSC JE Previous Year Questions — Centrifugal Pumps — Classification, Principles & Performance

Q1. Before starting, a centrifugal pump is:

SSC JE (Previous Year)

(a) Lubricated
(b) Primed (filled with water)
(c) Heated
(d) Emptied

✔ **Answer: (b) Primed (filled with water)**

Explanation: Priming.

Q2. Discharge of a pump varies with speed as:

SSC JE (Previous Year)

(a) Directly proportional
(b) Proportional to square
(c) Proportional to cube
(d) Inversely proportional

✔ **Answer: (a) Directly proportional**

Explanation: Q ∝ N.

Q3. A multi-stage centrifugal pump is used for:

SSC JE (Previous Year)

(a) High discharge
(b) High head
(c) Viscous liquids
(d) Low speed

✔ **Answer: (b) High head**

Explanation: Impellers in series.

Q4. Specific speed of a centrifugal pump depends on:

SSC JE (Previous Year)

(a) N, Q, H
(b) N, P, H
(c) Q, P
(d) Only N

✔ **Answer: (a) N, Q, H**

Explanation: $N\sqrt{Q}/H^{3/4}$.

Q5. Manometric head of a centrifugal pump is the:

SSC JE (Previous Year)

(a) Actual head developed by the pump

(b) Euler head

(c) Static head

(d) Velocity head

✔ **Answer: (a) Actual head developed by the pump**

Explanation: From suction and delivery gauges.

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